ARYAN AMIT TIWARI

+91 9971384597 | aryanamittiwari@gmail.com | Github | Portfolio | LinkedIn | LeetCode

Developer shipping applied AI and distributed systems. I build the backend infrastructure that powers autonomous agent workflows, quantitative data pipelines, and Web3 protocols, focusing on fast execution and clean, scalable code.

EXPERIENCE

- **VeLyra Labs(Stealth Startup)**
  Gen AI Intern 06/2026 - 10/2026
  - Engineered an autonomous multilingual AI agent capable of executing real-time transactional state mutations and role based escalations by integrating LLM workflows with a TypeScript monolith.
  - Architected a synchronization layer utilizing Redis pub/sub and WebSockets, resolving critical session eviction vulnerabilities to enable sub-100ms multi user concurrency.
  - Refined contextual data pipelines by expanding database queries to include granular metadata, eliminating hallucination loops and enforcing strict adherence to business constraints.
  - Managed secure production deployments via Azure DevOps, configuring KVS via RBAC and enforcing multi-tenant RLS across PostgreSQL database.
- **Wayspire(Microsoft & Tryst, IIT-Delhi)**
  AI Intern (Remote) 02/2024 - 03/2024
  - Incorporated and tested scalable NLP modules for intent classification, achieving accuracy by systematically analyzing model performance and refining algorithms.

PROJECTS

- **StakePort**
  - A blockchain platform for fractional ownership of high value assets using Web3 tokenization.
  - Dividing each asset into digital tokens that represent ownership shares, enabling easy buying, selling, and trading of asset fractions.
  - Leveraging a secure and transparent blockchain network to facilitate transactions.
  - Lowering investment barriers, improving liquidity, and access to valuable assets while ensuring security and transparency.
  - Implementing smart contracts to handle ownership transfers.
- **EcoLoop**
  - An autonomous multi-agent building controller that dynamic optimizes EnergyPlus HVAC simulations to reduce energy consumption while maintaining thermal comfort.
  - Engineered a decoupled architecture featuring an LLM cognitive engine for strategy, a deterministic supervisor for setpoint calculation, and a Guardian safety shield to enforce physical thermodynamic constraints.
  - Built fault-tolerant data pipelines with self-repair mechanisms for simulation stability and an interactive dashboard to monitor real-time power and comfort metrics.
  - Validated performance across extreme climate profiles (Chicago & Delhi) using a rigorous ablation ladder against traditional controllers.
- **MediVault**
  - A decentralized unified health ledger that grants patients absolute sovereign ownership of their medical history using Web3 architecture.
  - Leveraging a hybrid on/off-chain architecture, bridging a React and Node.js ecosystem with the Ethereum Sepolia network for secure data interoperability.
  - Eliminating centralized data silos and unauthorized breaches, while ensuring instant, permissioned access to vital records for verified medical practitioners.
  - Solidity smart contracts via Ethers.js to enforce granular access control.
- **Recon**
  - Building a fully autonomous, multi-agent AI system to simulate a quantitative hedge fund's research and execution pipeline.
  - Developing a specialized "crew" of AI agents that scrape financial data, debate risk, and formulate structured trading strategies using orchestration frameworks.
  - Integrating high-speed data pipelines, caching, and local NLP models to condense market noise and execute analysis.
  - Engineering an instant backtesting loop to validate generated strategies against historical data.

SKILLS

- Programming: Python, TypeScript, JavaScript, C++, C, SQL, Java
- AI & ML: LLM's, RAG, Autonomous Agents, Prompt Engineering, Fine Tuning, PyTorch, Deep Learning
- AI Frameworks: LangGraph, n8n LangChain, Hugging Face
- Databases: MongoDB, VectorDB, SQLite, PostgreSQL
- Cloud & DevOps: AWS, Docker, Azure, Git/GitHub, CI/CD
- Backend & APIs: FastAPI, Node.js, REST/GraphQL APIs, Web3/Blockchain, Smart Contracts, Cryptography

EDUCATION

VIT Chennai **2023 - 2027**
- BTech in CSE(AI & ML)

IIT Madras **2023 - Present**
- BS in Data Science

DPS Sushant Lok, Gurgaon
- Schooling

CERTIFICATIONS

NPTEL : EBusiness(IIT-K)
Business Fundamentals(IIT-B)
IBM: Gen AI
Finlatics: Business Analyst

INTERESTS

- Open Source Contributions
- Decentralized Systems
- Quantitative Finance & Algorithmic Trading